ARYAN BABURAJAN

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EXPERIENCE

Frappe.io
GitHub

Jan — Apr 2026
Intern

- Wrote a proof-of-concept Gunicorn worker using Python 3.14 Subinterpreters, enabling concurrent parallel execution beyond the limitations imposed by the GIL.
- Integrated Bloomberg's Memray profiler into Frappe's web dashboard to support memory and performance analysis.
- Extended Redis Queue's WorkerPool to allow workers to process multiple queues concurrently, improving memory usage by 32%.

PROJECTS

Ducktape
GitHub

- Built an open-source C++ 3D game engine with an Entity Component System architecture using EnTT, reaching 141+ GitHub stars and 28 forks.
- Implemented real-time rendering in OpenGL, with previous renderer implementations in Vulkan.
- Developed native C++ DLL-based scripting and an ImGui editor for in-engine debugging and content creation.

SerenityOS
GitHub

- Improved Unicode support across native applications, extending character handling beyond ASCII.
- Migrated applications to SerenityOS's GML markup language.

Mirage
GitHub

- Built a personalized product discovery platform with an authenticated dashboard tailored to users' interests.
- Implemented an email newsletter system that periodically delivers product recommendations using scheduled jobs.
- Built with Next.js, React, MongoDB, Kinde Auth, and Resend.

LLM Playground
GitHub

- Implemented the original "Attention Is All You Need" Transformer from scratch, including multi-head attention, sinusoidal positional embeddings, encoder-decoder layers, training, and text generation.

X-Embed
GitHub | Website

- Built a Twitter/X embedding service that generates screenshot-style previews from post URLs.

Quaviz
GitHub | Website

- Developed an interactive 3D visualizer for comparing the relative scale of objects, animals, landmarks, and vehicles.

ACHIEVEMENTS

- **1st Place (₹15K) — Google Developer Groups Physical AI Hackathon**
- **Game Development Workshops** — Held three workshops for 20+, 50+, and 100+ students on game development using p5.js, Unity, and Three.js.

EDUCATION

Christ College of Engineering
2025 — Present

Bachelor of Technology in Computer Science and Engineering
Kerala, India